$$\frac{d}{dx}(x^5) = 5x^4 \Rightarrow \int 5x^4\,dx = x^5$$

Similarly,

$$\frac{d}{dx}(x^7) = 7x^6 \Rightarrow \int 7x^6\,dx = x^7$$

$$\frac{d}{dx}(e^x) = e^x \Rightarrow \int e^x\,dx = e^x$$

$$\frac{d}{dx}(\log x) = \frac{1}{x} \Rightarrow \int \frac{1}{x}\,dx = \log x$$

and so on.



NOTE

While differentiating the constant term we get zero. But in the reverse process, that is on integration, unless you know the value of the constant we cannot include. That is why we include an arbitrary constant c to each integral value.

Therefore for the above examples, we usually write

$$\int 7x^6\,dx = x^7 + c$$

$$\int e^x\,dx = e^x + c$$

These integrals are also called improper integrals or indefinite integrals

Rules and some formulae on integration:

(i) $\int k\,dx = kx$

(ii) $\int x^n\,dx = \frac{x^{n+1}}{n+1}$

(iii) $\int e^x\,dx = e^x$

(iv) $\int \frac{1}{x}\,dx = \log x$

(v) $\int (u \pm v)\,dx = \int u dx \pm \int v dx$

Example 7.21

Integrate the following with respect to x.

(i) x^7 (ii) $\frac{1}{x^6}$ (iii) $\sqrt{x}$ (iv) $x^5 - 4x^2 + 3x + 2$